

Geometry & Electrical Performance Summary

The design is based on a wound roll of aluminum foil electrodes separated by paper dielectric, with a target capacitance of 10 μF and a foil width of 0.3 meters. The required physical dimensions change significantly based on the paper thickness and the composite dielectric constant ($\epsilon_r \approx 3.2$) of the oil-impregnated paper.

Parameter	Using 30 g/m ² Paper (~40 μm thick)	Using 60 g/m ² Paper (~75 μm thick)	Notes
Foil Length Required Approx. Outer Diameter	~47 meters ~6.3 cm	~88 meters ~11.2 cm	For one of the two foil strips. Estimated from winding volume around a 1 cm core.
Estimated Safe DC Voltage	300 - 400 V	550 - 750 V	Critical: This is a <i>practical safe voltage</i> , estimated at 20-30% of the material's theoretical limit to account for imperfections. Conservative estimate for well-impregnated, dry paper-oil composite. The oil displaces air and can increase breakdown strength significantly.
Material Breakdown Strength	Up to 30 kV/mm (theoretical)		

Preparation & Treatment Steps

The process is divided into three main phases, with the CaO treatment being a highly experimental and challenging addition.

1. Core Construction & Drying

- **Wind the Capacitor:** Assemble the dry, unimpregnated roll using the specified lengths of aluminum foil and paper.
- **Condition the Roll:** Bake the entire wound roll in an oven at 105-110°C for several hours to drive out absorbed moisture. Cool it in a dry environment.

2. Oil Treatment with CaO (Experimental & High-Risk)

- **Purpose:** To chemically bind residual water in the olive oil ($\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$) and neutralize free fatty acids.
- **Critical Challenge:** The reaction creates a fine, insoluble **Calcium Hydroxide** [Ca(OH)_2] powder suspended in the viscous oil, which is extremely difficult to remove completely.
- **Recommended Separation Method: Vacuum Filtration** with a **filter aid** (like diatomaceous earth) is the most effective practical method. Simple decantation will not work, and centrifugation may be insufficient.
- **Major Risk:** Any remaining Ca(OH)_2 in the oil can dissociate into conductive ions, severely degrading the dielectric's insulating properties and likely ruining the capacitor.

3. Final Impregnation & Sealing

- Immediately after drying, submerge the capacitor roll in the treated, warm olive oil in a vacuum chamber (or as close to a vacuum as possible) to ensure the oil fully penetrates the paper layers.
- Seal the impregnated capacitor in an airtight container to prevent moisture re-absorption.

Key Takeaways & Recommendations

- **Trade-off:** Thicker paper (60 g/m²) allows for a much higher voltage rating but results in a capacitor that is nearly twice as large.
- **Voltage is Theoretical:** The calculated safe voltages are optimistic estimates. Real-world performance depends entirely on the perfection of your drying, impregnation, and sealing.
- **CaO is Problematic:** The CaO treatment step introduces significant complexity and risk of failure. For a more reliable project, the recommended alternatives are to use a **non-contact "tea bag" method** with silica gel (desiccant) and calcium carbonate (acid scavenger), or to use a purpose-made dielectric fluid like transformer oil.

Glass pipe: https://webaruhaz.lombik.hu/uvegcsovek?product_id=1785